

Functions and Function Notation

ANSWER KEY



Function notation

- 1 Consider the function $f(x) = 2x^2 - 5x + 3$. Evaluate:
- a** $f(0) = 3$ **b** $f(2) = 1$ **c** $f(-1) = 10$ **d** $f\left(\frac{1}{2}\right) = 1$
- 2 Consider the function $g(x) = \sqrt{x+5}$. Evaluate:
- a** $g(4) = 3$ **b** $g(-1) = 2$ **c** $g(11) = 4$ **d** $g(-5) = 0$
- 3 Consider the function $h(x) = \frac{x-3}{x+2}$. Evaluate:
- a** $h(0) = -\frac{3}{2}$ **b** $h(3) = 0$ **c** $h(-1) = -4$ **d** $h(5) = \frac{2}{7}$
- 4 If $f(x) = 3x - 7$ and $f(a) = 11$, find the value of a .
 $3a - 7 = 11$, so $3a = 18$ and $a = 6$.
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Identifying functions

- 5 Determine whether each equation defines y as a function of x :
- a** $y = 2x + 1$ Yes — one output for each input. **b** $x = y^2$ No — e.g. $x = 4$ gives $y = 2$ or $y = -2$.
c $y = |x|$ Yes — one output for each input. **d** $x^2 + y^2 = 9$ No — a circle fails the vertical line test.
- 6 Explain how the vertical line test is used to decide whether a graph represents a function.
If every vertical line crosses the graph at most once, each input x has at most one output y , so the graph represents a function. If any vertical line crosses the graph more than once, it is not a function.
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Combining functions

- 7 Let $f(x) = x^2 + 1$ and $g(x) = 2x - 3$. Find:
- a** $(f+g)(x) = x^2 + 2x - 2$ **b** $(f-g)(x) = x^2 - 2x + 4$
c $(f \cdot g)(2) = 5 \cdot 1 = 5$ **d** $(f \circ g)(1) = f(-1) = 2$

- 8 A tutoring center charges a one-time registration fee plus an hourly rate, modeled by $C(n) = 250 + 45n$ riyals for n hours of tutoring.
- a Interpret the value $C(0)$ in context. $C(0) = 250$: the registration fee alone, before any tutoring hours.
 - b Find $C(12)$. $C(12) = 250 + 540 = 790$ riyals.
 - c Solve $C(n) = 925$ and interpret your answer. $45n = 675$, so $n = 15$: a total cost of 925 riyals buys 15 hours of tutoring.